

Evaluation of a behavioral assessment tool for dogs relinquished to shelters



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ABSTRACT

The goal of the present study was to evaluate a shortened, 42-item version of the *Canine Behavioral Assessment and Research Questionnaire* (C-BARQ^(S)) as a behavioral screening tool for dogs relinquished to animal shelters. In contrast to a previous finding, the current study found no consistent evidence that relinquishing owners gave unreliable or biased responses to the questionnaire depending on whether or not they believed that this information would be shared with shelter staff or used to evaluate dogs for adoption. Relinquishing owners' C-BARQ^(S) responses for items related to aggression and fear directed toward humans and other dogs correlated with independent subjective assessments of aggressiveness made by shelter staff (generalized linear mixed models, $P < 0.05$, $N = 84$ aggressive dogs, $N = 156$ non-aggressive dogs). In addition, C-BARQ^(S) scores successfully discriminated between dogs based on their eventual outcomes (i.e., adoption or euthanasia; Generalized Linear Mixed Models, $P < 0.05$, $N = 181$ adopted, $N = 177$ euthanized), indicating convergent validity. Follow-up surveys with a subset of adoptive owners ($N = 53$) revealed significant correlations (after correction for multiple comparisons) between relinquishing and adoptive owners' C-BARQ^(S) ratings for 3 behavioral traits: stranger-directed aggression ($r_s = 0.494$, $P < 0.001$), chewing inappropriate objects ($r_s = 0.402$, $P < 0.01$), and urination when left alone ($r_s = 0.421$, $P < 0.01$). Overall, the findings confirmed the value of this type of shelter intake survey instrument for screening owner-surrendered dogs for the presence of behavior problems.

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1. Introduction

Current evidence suggests that behavioral problems are probably the leading risk factor for relinquishment of dogs to animal shelters, with nearly 50% of relinquishing owners citing behavioral problems as a contributory factor,

and roughly a quarter citing them as the primary reason for relinquishment (Arkow and Dow, 1984; Herron et al., 2007; Miller et al., 1996; Patronek et al., 1996; Salman et al., 1998, 2000). Canine behavior problems are also a significant concern for animal shelters and adoption centers since behaviorally problematic dogs may be returned post-adoption, or pose a danger to shelter staff, adopting owners, and/or other members of the public and their animals (Christensen et al., 2007; Mondelli et al., 2004; Shore, 2005; Stephen and Ledger, 2006; Wells and Hepper, 2000). Furthermore, growing public pressure to improve the welfare and save the lives of shelter animals has created a perceived need for reliable and accurate methods

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of screening dogs for the presence of major behavior problems at the time of relinquishment (Bollen and Horowitz, 2008; Dowling-Guyer et al., 2011; Segurson et al., 2005).

Many shelters have responded to this need by conducting behavioral assessments on in-coming dogs as a means of evaluating their suitability for adoption or behavioral intervention (Bollen and Horowitz, 2008; Diesel et al., 2008; Dowling-Guyer et al., 2011; Mornement et al., 2010; Reid et al., 2004; Sternberg, 2002). Although several shelter-intake temperament tests have been developed, most are relatively labor-intensive and only limited published information is available regarding their reliability or their ability to predict behavior in the post-adoption home environment (Christensen et al., 2007; Hennessy et al., 2001; Mornement et al., 2010; Van der Borg et al., 1991). Despite the limitations of such tests, few shelters attempt to obtain behavioral information directly from relinquishing owners due to concerns that owners will provide false or unreliable information resulting from either a lack of interest, as a way to justify surrendering a well-behaved pet, or out of a desire to improve their pet's chances of avoiding euthanasia (DiGiacomo et al., 1998; Marston et al., 2004; Segurson et al., 2005; Stephen and Ledger, 2006).

The *Canine Behavioral Assessment and Research Questionnaire* (C-BARQ) was developed as a reliable standardized method for evaluating dogs for the presence and severity of behavioral problems (Hsu and Serpell, 2003; Duffy et al., 2008; Duffy and Serpell, 2012). Its potential use as a behavioral screening tool for dogs relinquished to animal shelters was investigated by Segurson et al. (2005) who employed trained assistants to administer the survey to a sample of 54 owners relinquishing dogs to a shelter in central California. The study found evidence that relinquishing owners tended to under-report certain behavior problems if they believed the information would be used by shelter staff to match dogs to appropriate homes. It was concluded that these owners intentionally understated some of their dogs' behavior problems in order to improve their chances of being adopted (Segurson et al., 2005). The average length of time it took to complete the version of the C-BARQ used in Segurson et al.'s study (12.2 min) also appeared to deter many owners from completing it voluntarily at the time of relinquishment.

The goal of the present study was to evaluate a shortened version of the C-BARQ as a behavioral screening tool for dogs relinquished to animal shelters. The theory that owners intentionally under-report dogs' behavior problems to improve their chances of adoption was tested by having participants read one of three distinct introductory preambles, each giving a different account of how the questionnaire might be used by shelter staff to determine the fates of relinquished dogs. Convergent validity of the modified instrument was investigated by examining: (1) its ability to discriminate between relinquished dogs based on their final dispositions (e.g. euthanasia versus placed for adoption), and (2) the level of association between the relinquishing owners' C-BARQ assessments and shelter personnel's subjective behavioral assessments of the same dogs for aggressiveness. The correlation between the modified C-BARQ scores of surrendered dogs and their behavior

approximately 2 months post-adoption was also investigated.

2. Methods

2.1. Questionnaire development

The C-BARQ is a widely used behavioral assessment tool for dogs with established reliability characteristics and demonstrated construct validity (Hsu and Serpell, 2003; Van den Berg et al., 2010; Duffy and Serpell, 2012; Jakuba et al., 2013). It comprises 14 subscales extracted by factor analysis, and an additional 22 miscellaneous items that are scored either in terms of severity or frequency. All C-BARQ items are presented as 5-point ordinal rating scales with a score of "0" corresponding to "absent/never" and a score of 4 corresponding to "severe/always." For the purposes of analysis, C-BARQ factor scores are calculated by averaging all the item scores within each factor (Hsu and Serpell, 2003; Duffy and Serpell, 2012; <http://www.cbarq.org>).

To develop a shortened version of the instrument, an existing dataset containing scores from a large population of both 'normal' and shelter-relinquished dogs was used. Selected questionnaire items were removed in a step-wise fashion from the standard C-BARQ factors, and Cronbach's alpha values calculated to determine the impact of each item on the internal reliability of the truncated factors. Since basing this selection of items on the strength of their original factor loadings would have resulted in the loss of certain items (e.g. separation-related destructiveness) that are thought to be important to owners' relinquishment decisions (Salman et al., 2000), the decision was made to retain these items while also maintaining alpha values above 0.7 wherever possible. Through this iterative process, the original or "standard" version of the C-BARQ was reduced from 100 to 42 items. This shortened version comprises 14 subscales and 9 miscellaneous items (generating a total of 23 scores), and is henceforth referred to as the C-BARQ^(S).¹ Preliminary trials suggested that the C-BARQ^(S) could typically be completed in <5 min.

Three distinct, printed versions of the C-BARQ^(S) were prepared, each with a different introductory preamble. The first of these stated that participants' responses might be used by shelter personnel when evaluating the dog for possible adoption (Non-Confidential group). The second stated that the responses would, "NOT be viewed by shelter employees" (Confidential group), and the third gave no statement regarding how the information might or might not be used or shared (Control group). To satisfy Institutional Review Board requirements on the use of human subjects for research, a short debriefing statement was included on a separate page at the end of the Non-Confidential version of the survey informing these participants that the preamble statement indicating that questionnaire results would be shared with shelter staff was untrue, and that all survey information would be treated confidentially. To reduce the likelihood that

¹ A copy of the standard C-BARQ was published by Duffy and Serpell (2012).

these participants might read the debriefing statement before completing their behavioral assessment, a message appeared at the top of the final page stating: “If you have not yet completed the questionnaire, please go back and answer all questions before proceeding to the statement at the bottom of this page.”²

To simulate the conditions under which such questionnaires would likely be administered in practice, the modified C-BARQ^(S) forms were distributed by the staff/receptionists at participating shelters at the time of each dog's relinquishment.

Prior to completing the 42-item behavioral assessment, participants were requested to provide information on the dog's name, age, breed, sex, approximate weight, age when acquired, where acquired from, whether spayed/neutered, and age when spayed/neutered (if known).

2.2. Participants and participant recruitment

All participants were recruited via one of three different animal shelters in the USA: a relatively well-funded, private, urban shelter located on the West Coast (Shelter A); a poorly funded, rural, municipal shelter located in a Western Mountain state (Shelter B), and a well-funded, private shelter located in a mid-Western city (Shelter C). Shelters were recruited initially via an announcement distributed via email to senior staff at >50 different US shelters. Fourteen shelters responded positively to the initial announcement. Four of these shelters were then selected to represent four US regions: West Coast, Western Mountain, Midwest and East Coast. Due to unforeseeable circumstances, the East Coast shelter dropped out of the study before data collection commenced. Two potential replacement shelters were then identified. Of these, one was unable to obtain final approval from its board to participate, and the other agreed to participate but never actually began collecting data. Time constraints prevented further efforts to recruit additional shelters, so the decision was made to proceed with the 3 original shelters. Each participating shelter was visited by one of the authors (KK) and provided with an instructional seminar and a written protocol on how to recruit subjects for participation in the study.³

Two different categories of individual were recruited for the study. The first consisted of dog owners who were relinquishing a dog to one of the participating shelters. The acceptance criteria for this group were that the relinquished dog was at least 6 months old, that it had been owned by the relinquishing person for at least 2 months, that it was not being surrendered for immediate euthanasia due to ill-health or old age, and that the relinquishing person could read English at a level that would enable them to complete the survey independently or without the assistance of shelter staff or volunteers. Although participating shelters were instructed to distribute copies of the C-BARQ^(S) to all relinquishing owners who met the

acceptance criteria at the time of relinquishment, surveys were completed voluntarily and it was not possible for shelter staff to keep reliable track of the proportion of relinquishing owners who actually completed questionnaires. Response rates to the C-BARQ^(S) are therefore unknown. The surveys were distributed over a period of 12 months with one of the three different introductory preambles (described above) in a random order, and owners were instructed to seal the completed questionnaires in pre-paid envelopes addressed to the study authors. Shelters were asked to collect these envelopes without opening them, and mail them to the authors at frequent intervals.

The second category of participants comprised individuals who were adopting previously relinquished dogs from one of the participating shelters. For an adoptive owner to be accepted for participation in the study, s/he needed to sign a form giving the authors formal permission to contact him/her approximately two months following adoption, and the adopted dog needed to have a completed C-BARQ^(S) on file from the previous (relinquishing) owner. Approximately two months post-adoption, these individuals were either sent the ‘standard’ (100 item) version of the C-BARQ to complete via email, or were invited to complete it online at: <http://www.cbarq.org>. Completed surveys were obtained from 53 adopting owners who acquired dogs from the three participating shelters (shelter A: $N=42$, shelter B: $N=1$, shelter C: $N=10$).

2.3. Correspondence with original C-BARQ and internal reliability

To determine the degree of correspondence between the subscale scores of the long (100-item) and short (42-item) versions of the C-BARQ, we used a convenience sample of currently-owned pet dogs ($N=17,307$) derived from the C-BARQ database at the University of Pennsylvania. Two sets of subscales scores were calculated from this sample of dogs: the original subscales and the subscales calculated using only the items included in the C-BARQ^(S).

To assess the internal reliability of the C-BARQ^(S) subscales, Cronbach's alpha was calculated using data from the three participating shelters to best represent the target population for this behavioral instrument.

2.4. Cross-validation of the C-BARQ^(S)

At shelters B and C, experienced staff routinely performs subjective behavioral intake assessments for aggressive behavior among relinquished dogs, during which each dog's aggressive responses to various targets (e.g., humans, other dogs) are recorded. These intake assessments provided an opportunity to evaluate the convergent validity of the C-BARQ^(S) by examining the degree of association between the results of these intake tests and the C-BARQ^(S) scores derived from relinquishing owners. To this end, a new binary variable – ‘aggression on intake’ – was created, and all dogs with any records of showing signs of aggression during the intake assessment, including biting and/or resource guarding (with or without signs of fear) either toward humans ($N=28$), other dogs ($N=11$) or no specific target ($N=49$), were scored as “1” on this variable, and all

² A copy of this version of the C-BARQ^(S) including the debriefing statement is available as supplementary material.

³ Copies of the written protocol are available from the corresponding author on request.

dogs showing no signs of aggression were scored as “0” ($N=84$ and $N=156$, respectively). These two groups of dogs were then compared for their C-BARQ^(S) scores.

In addition, we investigated whether relinquishing owners' C-BARQ^(S) scores were able to discriminate between relinquished dogs based on their final disposition (i.e., adopted versus euthanized) using data from all three shelters combined ($N=181$ and $N=177$, for adopted versus euthanized dogs, respectively). Outcomes not included in this analysis were ‘transferred’ ($N=48$) and ‘reclaimed’ ($N=27$).

2.5. Statistical analyses

All analyses were performed using SPSS version 21. C-BARQ^(S) scores were examined for normality using Q–Q plots. Five subscales met the assumption of normality (Excitability, Attachment/attention-seeking, Trainability, Chasing, and Energy). Box Cox transformations were applied to the remaining C-BARQ^(S) scores and the transformed scores were reexamined for normality. Box Cox transformations were performed as follows:

$$y_i^{(\lambda)} = \begin{cases} \frac{y_i^\lambda - 1}{\lambda} & \text{if } \lambda \neq 0, \\ \ln(y_i) & \text{if } \lambda = 0, \end{cases}$$

for which 30 values of lambda (λ), ranging from -2 to 1 in 0.1 increments, were tested (Osborne, 2010). Due to the presence of zeros in the data, a constant term (1) was added to the scores prior to transformation. Seven C-BARQ^(S) subscales were successfully transformed to meet the assumption of normality (stranger-directed aggression, stranger-directed fear, dog-directed aggression, nonsocial fear, dog-directed fear, touch sensitivity, and separation-related problems). The transformed variables were used in analysis then the estimated marginal means were back-transformed. The remaining C-BARQ^(S) subscales and items that did not meet the assumption of normality after transformation were converted into binary scores (zero and any score greater than zero; Table 1).

To determine whether the perceived ‘confidentiality’ (or lack thereof) of the C-BARQ^(S) influenced or biased relinquishing owners' responses to the questionnaire, generalized linear modeling (GLM) was used with C-BARQ^(S) subscale or item scores serving as the dependent variables (separate analyses for each C-BARQ^(S) item/subscale) and ‘confidentiality’, ‘shelter’ and their interaction serving as independent variables. The models used were either linear with an identity link function (for normally distributed C-BARQ^(S) scores or Box Cox transformed scores) or logistic with a logit link function (for C-BARQ^(S) scores that were transformed into binary variables).

Spearman's rank order correlations were used to compare rankings of dogs' scores on the C-BARQ subscales between relinquishing and adopting owners, and subscale scores calculated using the 100-item and 42-item versions of the C-BARQ using the sample of currently owned pets. Cronbach's alpha was used to assess the internal validity of the C-BARQ^(S) subscales using data collected from the three participating shelters.

C-BARQ^(S) scores of dogs that were and were not determined to be aggressive based on evaluations by shelter staff were compared using separate generalized linear mixed models (GLMM, binary with logit link function) for each C-BARQ^(S) trait, with ‘aggression on intake’ as the dependent variable and C-BARQ^(S) score, sex, age (centered around the mean), and shelter as fixed variables and breed as a random variable. The same GLMM analysis was used to compare the C-BARQ^(S) scores of dogs that were adopted versus those that were euthanized.

All tests were considered statistically significant at the 95% confidence level. Corrections for multiple comparisons were made using the Benjamini–Hochberg step-up FDR-controlling procedure (Benjamini and Hochberg, 1995).

2.6. Sample size estimation

Sample size estimates were based on previously published data (Segurson et al., 2005) and a convenience sample of unpublished data collected at the Veterinary Hospital of the University of Pennsylvania (VHUP). The raw data from the Segurson et al. (2005) study were used to calculate effect sizes for the C-BARQ items whose scores were found to differ significantly based on whether or not relinquishing owners believed their responses were to be kept confidential (owner-directed aggression and stranger-directed fear). Effect sizes for owner-directed aggression and stranger-directed fear were calculated using the subscales as continuous variables as well as dichotomized versions, where any score greater than zero was given a value of “1” and zeros remained “0”.

Using the Segurson et al. (2005) data, separate General Linear Models (GLM) with ‘confidentiality’, ‘shelter’, and their interaction as predictors and the C-BARQ score as the dependent variable were performed for owner-directed aggression and stranger-directed fear, yielding a partial Eta-squared value as an indicator of effect size ($\eta^2=0.06$ for Owner-directed aggression (continuous); 0.03 for Owner-directed aggression (binary); 0.05 for stranger-directed fear (continuous); and 0.06 for stranger-directed fear (binary)).

The effect sizes calculated using the Segurson et al. (2005) data were then used to estimate sample sizes needed for the current study. Sample size estimates were computed using the G*Power statistical power analysis program (version 3.1.9.2; Faul et al., 2007, 2009). In addition to the calculated effect sizes, other parameters for the sample size estimation were as follows: α error probability = 0.05 , Power ($1 - \beta$ err prob) = 0.95 , numerator df = 2 (based on three levels within the predictor variable of interest, confidentiality), and number of groups = 9 (three levels of confidentiality multiplied by three different shelters). The sample sizes estimated to achieve Power = 0.9 with $\alpha=0.05$ ranged from a total of 201 (67 per shelter with 22–23 in each ‘confidentiality’ group) to 374 (126 per shelter with 41–42 in each ‘confidentiality’ group).

Estimates of sample sizes needed for comparisons of the C-BARQ scores of relinquishing versus adopting owners were based on a convenience sample of pairs of dog owners living in the same households ($N=75$ pairs) recruited from

among former clients of the Veterinary Hospital of the University of Pennsylvania (VHUP). Each pair of owners was asked to complete a standard C-BARQ assessment of the same dog independently, and Spearman rank order correlations between the paired scores were calculated. Correlation coefficients ranged from 0.24 (Excitability) to 0.85 (stranger-directed aggression).

Using the G*Power statistical power analysis program (bivariate correlation model), the correlation coefficients from these data were then used to estimate the sample sizes needed for the current study for comparing C-BARQ scores from relinquishing owners to those from adopting owners. In addition to the correlation coefficients, other parameters used were as follows: one-tailed test, α error probability = 0.05, and Power ($1 - \beta$ err prob) = 0.95. The sample sizes estimated to achieve Power = 0.9 ranged from <10 to 32.

3. Results

3.1. The sample

Questionnaires were completed correctly by a total of 443 relinquishing owners from the three participating shelters; five of these did not meet eligibility criteria because the dogs were <6 months of age, leaving a final sample size of 438. The distribution of participants across the three different shelters was as follows: shelter A: $N=198$, shelter B: $N=85$, shelter C: $N=155$. Sixty-six breeds of purebred dogs were represented in the sample, as well as six categories of mixed breed dogs. The most

common breed categories included Labrador Retrievers and Labrador mixes ($N=87$, 19.9%); Pit Bulls and Pit Bull mixes ($N=62$, 14.2%); German shepherds and German shepherd mixes ($N=39$, 8.9%); and mixed breed/breed unknown ($N=38$, 8.7%). Fifty-five per cent of the sample comprised male dogs, and 57% of males and 60% of females were neutered/spayed. The dogs' reported ages ranged from 0.5 to 14 years (mean $\pm$ S.D. = 3.73 ± 3.04).

3.2. Correspondence with original C-BARQ and internal reliability

Internal reliability of the C-BARQ^(S) subscales were calculated using the data collected from the three shelters ($N=438$). All subscales had Cronbach's alpha scores of ≥ 0.70 with the exception of 'nonsocial fear' ($\alpha=0.625$) and 'trainability' ($\alpha=0.504$). Removal of one item from the 'trainability' subscale ("easily distracted by interesting sights, sounds or smells") improved the Cronbach's alpha score to 0.742; therefore, this item was excluded from the calculation of the 'trainability' subscale scores.

Subscales calculated using only the items that appeared in the C-BARQ^(S) were highly correlated with subscales calculated using the original 100-item C-BARQ (currently owned pet data, $N=17,307$; Spearman's rank order correlation, $r_s \geq 0.80$, $P < 0.0001$ for all subscales).

3.3. Cross-validation of the C-BARQ^(S)

Dogs that were recorded by shelter staff as showing aggressive behavior during intake evaluations (i.e., scored 1

Table 1

Results of generalized linear mixed models examining the relationship between the C-BARQ^(S) trait scores and the designation of dogs as either aggressive or non-aggressive by shelter staff.

C-BARQ trait (independent variable)*	N**	Transform	Coeff.	95% CI	P value
<i>Subscales</i>					
Excitability	229	None	0.28	−0.01, 1.32	0.059
Attachment/attention-seeking	234	None	0.01	−0.26, 0.28	0.941
Trainability	235	None	0.19	−0.17, 0.55	0.305
Chasing	233	None	−0.08	0.30, 0.13	0.452
Energy	237	None	−0.02	−0.28, 0.24	0.861
Stranger-directed aggression	234	Box Cox	1.37	−0.62, 2.13	0.0004
Dog-directed aggression	224	Box Cox	0.70	0.17, 1.22	0.009
Stranger-directed fear	226	Box Cox	1.42	0.45, 2.40	0.004
Dog-directed fear	225	Box Cox	0.89	0.33, 1.45	0.002
Nonsocial fear	232	Box Cox	0.32	−0.30, 0.93	0.310
Touch sensitivity	207	Box Cox	0.23	−0.97, 0.51	0.543
Separation-related problems	231	Box Cox	0.77	0.12, 1.43	0.021
Owner-directed aggression	233	Binary	0.87	0.29, 1.44	0.003
Dog rivalry	190	Binary	0.91	0.27, 1.56	0.006
<i>Miscellaneous items</i>					
Escapes from house/yard	236	Binary	−0.12	−0.68, 0.45	0.684
Chews on inappropriate objects	235	Binary	−0.46	−1.06, 0.43	0.135
Pulls on leash	234	Binary	0.52	−0.22, 1.25	0.167
Urine marking	234	Binary	0.02	−0.58, 0.62	0.953
Urination when left alone	232	Binary	−0.60	−1.19, −0.02	0.044
Defecation when left alone	230	Binary	−0.47	−1.06, 0.12	0.120
Hyperactive	237	Binary	0.45	−0.13, 1.02	0.128
Tail chasing	236	Binary	0.80	0.24, 2.23	0.006
Persistent barking	236	Binary	0.44	−0.24, 1.13	0.203

Bold font indicates significant effect with Benjamini–Hochberg step-up FDR-controlling procedure.

* All models included shelter, age (centered around the mean), and sex as fixed effects and breed as a random effect (all non-significant).

** Only 2 of 3 shelters provided information about aggression during evaluation upon intake.

on 'aggression on intake') obtained significantly less favorable C-BARQ^(S) scores on all four aggression subscales, in addition to several other behavioral variables (Table 1 and Fig. 1; GLMM, $P < 0.05$ with correction for multiple comparisons).

Compared with the euthanized dogs, dogs offered for adoption obtained significantly more favorable C-BARQ^(S) scores on "stranger-directed aggression" (Table 2; GLMM, $P < 0.05$ with correction for multiple comparisons).

3.4. Did relinquishing owners under-report behavior problems?

The information that relinquishing owners were given regarding how the survey results might be used (e.g. confidential, non-confidential, or control) had no consistent impact on how they described their dogs' behavior using the C-BARQ^(S) (Table 3). With correction for multiple comparisons, only a main effect for 'shelter' was significant at the $P < 0.05$ level. No main effects for 'confidentiality' were found, with or without correction for multiple comparisons.

3.5. Pre- and post-adoption behavioral assessments

Comparisons between initial relinquishment C-BARQ^(S) scores and those provided by adopting owners at least two months post-adoption using Spearman rank order correlation coefficients revealed only 3 positive correlations between relinquishing and adopting owners' C-BARQ assessments after correction for multiple comparisons (Table 4): stranger-directed aggression ($r_s = 0.494$,

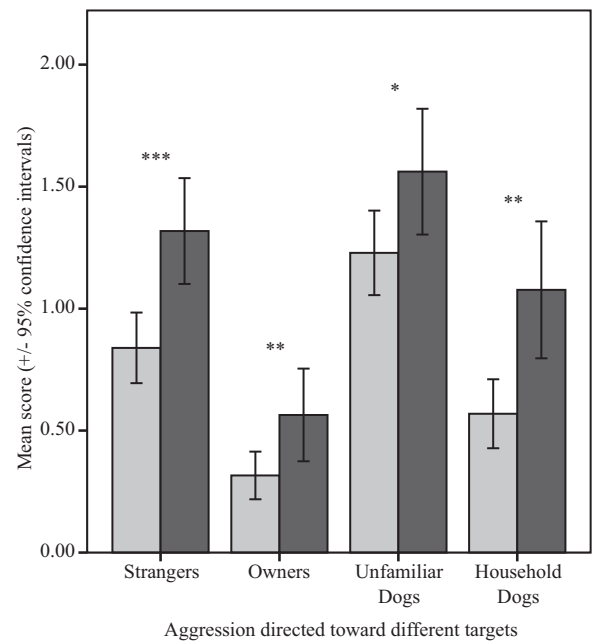


Fig. 1. Mean ($\pm 95\%$ CI) C-BARQ^(S) scores of dogs identified by shelter staff as aggressive on intake (dark gray bars) compared to dogs not determined to be aggressive on intake (light gray bars) for the four subscales "stranger-directed aggression," "owner-directed aggression," "dog-directed aggression" and "familiar/household dog aggression". The possible scale range for all C-BARQ^(S) subscales is from 0 to 4. * $P < 0.01$, ** $P < 0.001$, *** $P < 0.0001$.

Table 2

Results of generalized linear mixed models examining the relationship between the C-BARQ traits and outcomes of dogs (euthanized or adopted).

C-BARQ trait (independent variable) ^a	N	Transform	Coeff.	95% CI	P value
Subscales					
Excitability	350	None	0.30	0.06, 0.55	0.016
Attachment/attention-seeking	350	None	−0.04	−0.26, 0.19	0.748
Trainability	351	None	0.26	−0.06, 0.58	0.107
Chasing	344	None	0.10	−0.09, 0.28	0.301
Energy	355	None	0.08	−0.15, 0.32	0.485
Stranger-directed aggression	353	Box Cox	1.01	0.39, 1.62	0.002
Dog-directed aggression	338	Box Cox	0.62	0.19, 1.05	0.005
Stranger-directed fear	344	Box Cox	0.90	0.10, 1.70	0.027
Dog-directed fear	336	Box Cox	0.41	−0.64, 0.88	0.090
Nonsocial fear	348	Box Cox	0.47	−0.05, 1.00	0.078
Touch sensitivity	287	Box Cox	−0.09	−0.73, 0.54	0.780
Separation-related problems	347	Box Cox	0.47	−0.06, 1.00	0.084
Owner-directed aggression	353	Binary	0.62	0.14, 1.10	0.012
Dog rivalry	267	Binary	0.45	−0.08, 0.98	0.096
Miscellaneous items					
Escapes from house/yard	350	Binary	0.25	−0.22, 0.73	0.289
Chews on inappropriate objects	355	Binary	0.02	−0.32, 0.64	0.518
Pulls on leash	349	Binary	0.55	−0.06, 1.16	0.076
Urine marking	353	Binary	0.17	−0.35, 0.68	0.528
Urination when left alone	346	Binary	−0.16	−0.62, 0.31	0.507
Defecation when left alone	343	Binary	−0.03	−0.51, 0.45	0.899
Hyperactive	354	Binary	0.01	−0.45, 0.47	0.960
Tail chasing	353	Binary	0.60	0.14, 1.06	0.011
Persistent barking	354	Binary	0.60	0.06, 1.15	0.031

Bold font indicates significant effect with Benjamini–Hochberg step-up FDR-controlling procedure.

^a All models included shelter, age (centered around the mean), and sex as fixed effects and breed as a random effect (all non-significant).

Table 3

Results of generalized linear models examining the relationship between the perceived confidentiality of the relinquishing owners' survey responses and C-BARQ scores.

Dependent variable ^a	N	Transform	Model	Sig. effects	Estimated marginal means								
					Shelter A			Shelter B			Shelter C		
					NC	C	CT	NC	C	CT	NC	C	CT
<i>Subscales</i>													
Excitability	427	None	1	n.s.	2.40	2.37	2.59	2.48	2.35	2.39	2.24	2.28	2.28
Attachment/attention-seeking	429	None	1	a	2.57	2.49	2.60	2.98	2.73	2.89	2.86	2.83	2.56
Trainability	431	None	1	a	2.42	2.42	2.42	2.41	2.44	2.15	2.24	2.16	2.20
Chasing	421	None	1	a	1.94	1.77	1.74	1.48	1.62	1.88	2.11	2.13	1.93
Energy	434	None	1	n.s.	2.54	2.70	2.58	2.95	2.67	2.64	2.82	2.70	2.59
Stranger-directed aggression	430	Box Cox ^{**}	1	n.s.	0.82	0.76	0.68	0.89	0.69	0.85	0.98	0.88	0.77
Dog-directed aggression	411	Box Cox ^{**}	1	n.s.	1.12	1.07	0.97	1.16	1.03	1.60	1.21	1.12	1.14
Stranger-directed fear	419	Box Cox ^{**}	1	a	0.50	0.45	0.34	0.55	0.69	0.69	0.64	0.43	0.40
Nonsocial fear	426	Box Cox ^{**}	1	n.s.	1.00	1.03	0.95	0.73	1.02	1.22	0.87	0.89	0.84
Dog-directed fear	411	Box Cox ^{**}	1	n.s.	1.07	0.96	1.05	0.95	0.81	1.43	1.12	0.97	0.99
Touch sensitivity	351	Box Cox ^{**}	1	n.s.	0.84	0.60	0.73	0.62	0.36	0.78	0.81	0.69	0.67
Separation-related problems	426	Box Cox ^{**}	1	n.s.	0.77	0.74	0.92	0.92	0.83	0.98	0.99	0.98	0.97
Owner-directed aggression	430	Binary	2	a ^{***} , b	0.59	0.86	0.80	0.64	0.59	0.57	0.53	0.52	0.53
Dog rivalry	332	Binary	2	n.s.	0.40	0.50	0.62	0.48	0.42	0.42	0.32	0.53	0.48
<i>Miscellaneous items</i>													
Escapes from house/yard	429	Binary	2	n.s.	0.68	0.66	0.68	0.43	0.81	0.71	0.62	0.65	0.65
Chews on inappropriate objects	432	Binary	2	n.s.	0.35	0.33	0.52	0.37	0.27	0.22	0.34	0.33	0.41
Pulls on leash	426	Binary	2	n.s.	0.15	0.16	0.10	0.21	0.15	0.19	0.20	0.19	0.23
Urine marking	430	Binary	2	n.s.	0.78	0.78	0.74	0.73	0.62	0.69	0.69	0.62	0.64
Urination when left alone	420	Binary	2	n.s.	0.60	0.50	0.66	0.73	0.75	0.54	0.53	0.56	0.53
Defecation when left alone	416	Binary	2	n.s.	0.72	0.73	0.67	0.73	0.62	0.69	0.65	0.61	0.55
Hyperactive	433	Binary	2	n.s.	0.31	0.38	0.52	0.30	0.38	0.39	0.41	0.42	0.44
Tail chasing	432	Binary	2	n.s.	0.61	0.66	0.60	0.57	0.46	0.54	0.59	0.56	0.62
Persistent barking	433	Binary	2	n.s.	0.26	0.22	0.15	0.17	0.19	0.07	0.18	0.35	0.33

NC, not confidential; C, confidential; CT, control; 1, linear with identity link function; 2, logistic with logit link function; a, main effect for shelter; b, interaction effect for shelter × confidentiality.

^a Confidentiality, shelter and their interaction were all included as fixed effects.

^{**} Estimated marginal means were back-transformed for variables that had been Box Cox transformed for analysis.

^{***} Remained significant at the $P < 0.05$ level with Benjamini–Hochberg step-up FDR-controlling procedure.

Table 4

Correlations between C-BARQ scores based on comparisons of relinquishing and adopting owners ($N = 53$).

Subscale or item	rho	P value
Escapes from house/yard	0.312	0.031
Chews on inappropriate objects	0.402	0.004
Pulls on leash	0.328	0.021
Urine marking	0.262	0.067
Urination when left alone	0.421	0.003
Defecation when left alone	0.110	0.468
Hyperactive	−0.049	0.731
Tail chasing	0.345	0.014
Persistent barking	0.324	0.020
Excitability	−0.065	0.656
Stranger-directed aggression	0.494	0.0004
Owner-directed aggression	0.227	0.117
Dog-directed aggression	0.319	0.031
Dog rivalry	0.072	0.785
Stranger-directed fear	0.007	0.962
Nonsocial fear	−0.180	0.226
Dog-directed fear	0.187	0.241
Touch sensitivity	0.291	0.159
Separation-related problems	0.009	0.948
Attachment/attention-seeking	0.262	0.069
Trainability	0.266	0.065
Chasing	−0.030	0.845
Energy	0.200	0.164

Bold font indicates significant effect with Benjamini–Hochberg step-up FDR-controlling procedure.

$P < 0.001$), chewing inappropriate objects ($r_s = 0.402$, $P < 0.01$), and urination when left alone ($r_s = 0.421$, $P < 0.01$).

4. Discussion

The primary goal of this study was to evaluate the C-BARQ^(S) as an assessment tool for screening dogs for behavior problems at the time of their relinquishment to animal shelters. The results suggest that this instrument provides a valid and reliable method for collecting behavioral information from relinquishing owners, although the exclusion of one item from the “trainability” section (“easily distracted by interesting sights, sounds or smells”) helped to improve the reliability of this subscale. Comparisons of the standard C-BARQ and C-BARQ^(S) subscales showed strong correlations, while Cronbach's alpha scores demonstrated adequate–good internal reliability for all subscales of the C-BARQ^(S) with the exception of “nonsocial fear”.

In contrast to the results of a previous study (Segurson et al., 2005), knowledge of how the survey information might or might not be used or shared with shelter staff had no significant impact on relinquishing owners' responses. Given that the current study included a substantially larger sample size compared to Segurson et al. ($N = 438$ versus $N = 54$), it is unlikely that the lack of a significant effect is due to insufficient statistical power. This suggests that the C-BARQ^(S) responses were relatively free from

systematic over- or under-reporting biases. The lack of systematic biases in the reporting of behavior problems was also consistent across the three different shelters that participated in the study. It is not known why the current findings differed from those of Segurson et al. (2005), although the latter's use of a smaller sample size may have increased the possibility of a Type I error (Martin and Bateson, 1993). Also, the use of face-to-face interviews by Segurson et al. (2005) may have rendered their findings more susceptible to social desirability biases – i.e., the tendency to respond to survey items in a manner that will be viewed more favorably by others rather than responding in an accurate and truthful manner. Previous studies have demonstrated that the results of self-administered questionnaires of the type used in the present study show fewer social desirability effects than those administered by face-to-face interview, probably because the former allow a greater degree of neutrality (Fisher, 1993; Kreuter et al., 2009). While it is possible that some participants who received the “non-confidential” version of the questionnaire in the present study read the debriefing statement on the final page before completing their behavioral assessment, and adjusted their responses to the C-BARQ^(S) accordingly, appropriate precautions were taken to reduce the likelihood of this happening.

The lack of systematic behavioral tests or observations by participating shelters at the time of dog intake made it difficult to assess the validity of the C-BARQ^(S) in the present study. Despite this limitation, the instrument's ability to discriminate between dogs based on the outcomes of independent, in-house behavioral assessments as well as decisions regarding adoption versus euthanasia tended to confirm the validity of the C-BARQ^(S), particularly with respect to aggressive behavior problems.

Pre- and post-adoption assessments of dogs were significantly correlated for only 3 C-BARQ^(S) variables after correction for multiple comparisons: aggression directed toward unfamiliar people, chewing inappropriate objects, and house-soiling (urination) when left alone. The fact that shelter staff had selected this subset of dogs for adoption on the basis of ‘good’ behavior – thereby skewing the distribution and reducing variation – may have masked or weakened some associations between pre- and post-adoption surveys. Nonetheless, the findings are somewhat similar to those of a previous questionnaire study that used shorter intervals between relinquishment and post-adoption follow-up (2 and 6 weeks versus 2 months) but also found that “aggression toward unfamiliar people” showed the strongest pre- and post-adoption correlation (Stephen and Ledger, 2006). The relatively low level of correspondence between reported behavior in relinquishing versus adopting households may reflect systematic response biases among relinquishing and/or adopting owners, or may be due to actual changes in the dogs' behavior between the two different situations. Further studies will be needed to resolve this question.

Some limitations to the study should be noted. Since participation by both shelters and relinquishing owners was voluntary, the samples were self-selected and may not have been representative of the relinquishing owner population as a whole. Segurson et al. (2005) reported a

54% response rate to face-to-face C-BARQ administration in their study of relinquishing owners, and it would be interesting to know whether self-report administration of a shorter questionnaire – as in the current study – produces better or worse response rates in this context. Unfortunately, the decision to allow the shelters to distribute the C-BARQ^(S) themselves made it impossible to reliably determine response rates in our study. Low response rates would likely be a matter of concern to shelters contemplating the use of such a questionnaire, although perhaps requiring questionnaire completion as a condition of dog relinquishment might be one way to overcome this problem in practice.

5. Conclusions

Taken as a whole, these findings suggest that standardized questionnaires, such as the C-BARQ^(S), can be used by shelters to collect valid and reliable behavioral information from people relinquishing dogs to shelters. Such instruments may provide helpful and inexpensive adjuncts to existing behavioral and temperament screening methods currently in use, as well as being potentially useful tools for assessing the behavioral progress of dogs post-adoption. The widespread adoption of such standardized behavioral assessments by shelters could also provide a valuable source of epidemiological data on the distribution, prevalence and severity of behavior problems in the relinquished dog population.

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Appendix A. Supplementary data

Supplementary data associated with this article can be found, in the online version, at <http://dx.doi.org/10.1016/j.prvetmed.2014.10.003>.

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